Q1. $\underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} \sim N_3\left(\underset{\sim}{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{pmatrix}, \ \Sigma = (1-\rho) I_3 + \rho \underset{\sim}{1}\underset{\sim}{1}^T\right)$

(a) $$\underset{\sim}{Z} = \begin{bmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \\ Y_1 + Y_2 + Y_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} \underset{\sim}{Y} = A\underset{\sim}{Y}, \text{ say.}$$

So, $\underset{\sim}{Z} \sim N_3\left(A\underset{\sim}{\mu}, A\Sigma A'\right)$ — (*)

where $A\underset{\sim}{\mu} = \begin{bmatrix} \mu_1 - \mu_2 \\ \mu_2 - \mu_3 \\ \mu_1 + \mu_2 + \mu_3 \end{bmatrix}$ and — (I)

$$A\Sigma A' = (1-\rho) AA' + \rho A\underset{\sim}{1} (A\underset{\sim}{1})' . \quad — (1)$$

Now, $$AA' = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \quad — (2)$$

and $$A\underset{\sim}{1} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \Rightarrow A\underset{\sim}{1}(A\underset{\sim}{1})^T = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 9 \end{pmatrix} \quad — (3)$$

From (1) - (3) we get: $$A\Sigma A' = \begin{bmatrix} 2(1-\rho) & -(1-\rho) & 0 \\ -(1-\rho) & 2(1-\rho) & 0 \\ 0 & 0 & 3+6\rho \end{bmatrix} \quad — (II)$$

Thus, (*), along with (I) and (II) completes the result.

(b) From the above distribution of $\underset{\sim}{Z} = \begin{bmatrix} Z_1 \\ Z_2 \\ Z_3 \end{bmatrix} = \begin{bmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \\ Y_1 + Y_2 + Y_3 \end{bmatrix}$, it is clear that $\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} \perp\!\!\!\perp Z_3$ (as they jointly follow normal distribution and $\mathrm{Cov}\left(\begin{pmatrix} Z_1 \\ Z_3 \end{pmatrix}, Z_3\right) = \underset{\sim}{0}$). Thus, $\begin{pmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \end{pmatrix} \perp\!\!\!\perp \left(\frac{Y_1 + Y_2 + Y_3}{3}\right) = \bar{Y}$.

So, the marginal distn. of $\begin{bmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \end{bmatrix}$ is same as ~~the~~ its conditional distribution given $\bar{Y}$.

From (a), the marginal of $\begin{pmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \end{pmatrix}$ is $N_2\left(\begin{pmatrix} \mu_1-\mu_2 \\ \mu_2-\mu_3 \end{pmatrix}, (1-\rho)\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\right)$.

[ANS]

(c) $\underset{\sim}{W} = \underset{\sim}{Y} + \underset{\sim}{a}$; $\underset{\sim}{a} \in \mathbb{R}^3$.

Define $\underset{\sim}{V} = \begin{bmatrix} \underset{\sim}{W} \\ Y_1+Y_2+Y_3 \end{bmatrix} = B\underset{\sim}{Y} + \underset{\sim}{a_0}$ where $\underset{4\times 3}{B} = \begin{bmatrix} I_3 \\ \underset{\sim}{1}' \end{bmatrix}$ and $\underset{\sim}{a_0} = \begin{bmatrix} \underset{\sim}{a} \\ 0 \end{bmatrix}$

Clearly, $\underset{\sim}{V} \sim N\left(B\underset{\sim}{\mu} + \underset{\sim}{a_0} = \begin{bmatrix} \mu_1+a_1 \\ \mu_2+a_2 \\ \mu_3+a_3 \\ \mu_1+\mu_2+\mu_3 \end{bmatrix}, B\Sigma B^T\right)$

where $B\Sigma B^T = \begin{bmatrix} \Sigma \\ \underset{\sim}{1}'\Sigma \end{bmatrix} \begin{bmatrix} I_3 & \underset{\sim}{1} \end{bmatrix} = \begin{bmatrix} \Sigma & \Sigma \underset{\sim}{1} \\ \underset{\sim}{1}'\Sigma & \underset{\sim}{1}'\Sigma\underset{\sim}{1} \end{bmatrix}$

and $\Sigma \underset{\sim}{1} = (1-\rho)\underset{\sim}{1} + \rho\underset{\sim}{1}\cdot 3 = (1+2\rho)\underset{\sim}{1}$;

$\underset{\sim}{1}'\Sigma\underset{\sim}{1} = 3(1+2\rho)$.

The conditional distribution of $\underset{\sim}{W}$ given $\underbrace{Y_1+Y_2+Y_3}_{=3\bar{y}}$ (equivalently, given $\underset{=\bar{y}}{\bar{Y}}$) is

$$N_3\left(\underset{\sim}{\mu_1} + \underset{\sim}{\sigma_{12}}(3\bar{y} - \mu_2^*)/\sigma_{22},\ \Sigma_{11} - \underset{\sim}{\sigma_{12}}\underset{\sim}{\sigma_{12}}^T/\sigma_{22}\right) \quad \text{—(A)}$$

where $\underset{\sim}{\mu_1} = \begin{pmatrix} \mu_1+a_1 \\ \mu_2+a_2 \\ \mu_3+a_3 \end{pmatrix} = \underset{\sim}{\mu} + \underset{\sim}{a}$; $\underset{\sim}{\sigma_{12}} = \Sigma\underset{\sim}{1} = (1+2\rho)\underset{\sim}{1}$,

$\Sigma_{11} = \Sigma$, $\sigma_{22} = \underset{\sim}{1}'\Sigma\underset{\sim}{1} = 3(1+2\rho)$, and $\mu_2^* = (\mu_1+\mu_2+\mu_3)$

Therefore $\underset{\sim}{\mu_1} + \underset{\sim}{\sigma_{21}}(3\bar{y} - \mu_2)/\sigma_{22} = \underset{\sim}{\mu} + \underset{\sim}{a} + \dfrac{(3\bar{y} - (\mu_1+\mu_2+\mu_3))}{3(1+2\rho)}(1+2\rho)\underset{\sim}{1}$

$$= \underset{\sim}{\mu} + \underset{\sim}{a} + (\bar{y} - \bar{\mu})\underset{\sim}{1} \quad \text{— (B)}$$

and $\Sigma_{11} - \underset{\sim}{\sigma_{12}}\underset{\sim}{\sigma_{12}}^T/\sigma_{22} = \Sigma - \dfrac{1}{3(1+2\rho)}(1+2\rho)^2\underset{\sim}{1}\underset{\sim}{1}'$

$$= (1-\rho)I + \left(\rho - \tfrac{1}{3}(1+2\rho)\right)\underset{\sim}{1}\underset{\sim}{1}'$$

$$= (1-\rho)I - \tfrac{1}{3}(1-\rho)\underset{\sim}{1}\underset{\sim}{1}' \quad \text{— (C)}$$

Combining Ⓐ - Ⓒ, we get:

The conditional distributn. of $\underset{\sim}{W}$ given $\bar{Y} = \bar{y}$ is:

$$N_3\left(\underset{\sim}{\mu} + \underset{\sim}{a} + (\bar{y} - \bar{\mu})\underset{\sim}{1} \;,\; (1-\rho) I - \frac{1}{3}(1-\rho)\underset{\sim}{1}\underset{\sim}{1}'\right)$$

[ANS]

Q2 Given $X_i \overset{ind}{\sim} \text{Gamma}(\alpha_i, 1)$ for $i = 1, \cdots, p$.

$$\begin{aligned} Y_1 &= X_1 Y_p^{-1} \\ &\vdots \\ Y_{p-1} &= X_{p-1} Y_p^{-1} \\ Y_p &= X_1 + \cdots + X_p \end{aligned} \quad \Rightarrow \quad \begin{aligned} X_1 &= Y_1 Y_p \\ &\vdots \\ X_{p-1} &= Y_{p-1} Y_p \\ X_p &= Y_p - (X_1 + \cdots + X_{p-1}) \\ &= Y_p \{1 - (Y_1 + \cdots + Y_{p-1})\}. \end{aligned}$$

(a) $\underset{\sim}{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_p \end{pmatrix} \to \begin{pmatrix} Y_1 \\ \vdots \\ Y_p \end{pmatrix} = \underset{\sim}{Y}$ is a one-one transformation.

Jacobian of transformation

$$J = \det\left(\left(\frac{\partial \underset{\sim}{X}}{\partial \underset{\sim}{Y}}\right)\right) = \det\left[\begin{bmatrix} \frac{\partial x_1}{\partial y_1} & \cdots & \frac{\partial x_1}{\partial y_p} \\ \vdots & & \\ \frac{\partial x_p}{\partial y_1} & \cdots & \frac{\partial x_p}{\partial y_p} \end{bmatrix}\right]$$

$$= \det\left[\underbrace{\begin{bmatrix} y_p & 0 & \cdots & 0 & y_1 \\ \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & y_p & y_{p-1} \\ -y_p & -y_p & \cdots & -y_p & (1 - y_1 - \cdots - y_{p-1}) \end{bmatrix}}_{M}\right]$$

Define the blocks $A = y_p I_{p-1}$; $B = \begin{bmatrix} y_1 \\ \vdots \\ y_{p-1} \end{bmatrix}$; $C = -y_p \underset{\sim}{1}'$ and $D = (1 - y_1 - \cdots - y_{p-1})$

Then $\det(M) = \det(A) \times \det(D - CA^{-1}B)$

$$= y_p^{p-1} \times \det\left((1 - y_1 - \cdots - y_{p-1}) + y_p y_p^{-1}(y_1 + \cdots + y_{p-1})\right) = y_p^{p-1}$$

Therefore, the joint pdf of $\underset{\sim}{Y}_p = \begin{pmatrix} y_1 \\ \vdots \\ y_p \end{pmatrix}$ is

$$f_{\underset{\sim}{Y}_p}(\underset{\sim}{y}_p) = f_{\underset{\sim}{X}}(\underset{\sim}{x}) \Big|_{\underset{\sim}{x} \to \underset{\sim}{y}} |J|$$

$$= \prod_{i=1}^{p} \left\{ \frac{1}{\Gamma\alpha_i} x_i^{\alpha_i - 1} e^{-x_i} \right\} \Big|_{\underset{\sim}{x} \to \underset{\sim}{y}} |J|$$

$$= \frac{1}{\prod_{j=1}^{p} \Gamma\alpha_i} e^{-(x_1 + \cdots + x_p)} \prod_{i=1}^{p} x_i^{\alpha_i - 1} \Big|_{\underset{\sim}{x} \to \underset{\sim}{y}} |J|$$

$$= \frac{1}{\prod_{i=1}^{p} \Gamma\alpha_i} e^{-y_p} y_1^{\alpha_1 - 1} \cdots y_{p-1}^{\alpha_{p-1} - 1} (1 - y_1 - \cdots - y_{p-1})^{\alpha_p - 1} \cdot y_p^{\alpha_1 + \cdots + \alpha_p - p} |J|$$

$$= \frac{1}{\prod_{i=1}^{p} \Gamma\alpha_i} e^{-y_p} y_p^{\alpha_1 + \cdots + \alpha_p - 1} y_1^{\alpha_1 - 1} \cdots y_{p-1}^{\alpha_{p-1} - 1} (1 - y_1 - \cdots - y_{p-1})^{\alpha_p - 1} ; \quad \text{— (I)}$$

Range: As $x_i > 0 \ \forall i$, $y_i \in (0,1)$ for $i = 1, \cdots, p-1$ and $y_p > 0$. } – (II)

From (I) and (II), we obtain the joint PDF of $\underset{\sim}{Y}_p$.

(b) From (I) it is clearly that $\underset{\sim}{Y}_{p-1} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_{p-1} \end{pmatrix}$ is independent of Y_p (as the joint PDF can be written as a product of two functions $g(\underset{\sim}{y}_{p-1})$ and $h(y_p)$]

Thus, the marginal of Y_p is proportional to $e^{-y_p} y_p^{\alpha_1 + \cdots + \alpha_p - 1}$, which is proportional to Gamma $(\alpha_1 + \cdots + \alpha_p, 1)$ PDF.

So, $Y_p \sim$ Gamma $(\alpha_1 + \cdots + \alpha_p, 1)$.

Therefore, the joint PDF of $\underset{\sim}{Y}_{p-1}$ is:

$$f_{\underset{\sim}{Y}_{p-1}}(\underset{\sim}{y}_{p-1}) = \frac{\Gamma\alpha_1 + \cdots + \alpha_p}{\Gamma\alpha_1 \cdots \Gamma\alpha_p} y_1^{\alpha_1 - 1} \cdots y_{p-1}^{\alpha_{p-1} - 1} (1 - y_1 - \cdots - y_{p-1})^{\alpha_p - 1} ; \quad y_i \in (0,1), \ i = 1, \cdots, p-1.$$